

An Access Network is the part of a telecommunication network that connects end users (homes, offices, mobile devices) to the Internet Service Provider (ISP) or core network.

CT Access Network

An **access network** is a type of network that physically connects an end system to the immediate router (also known as the “edge router”) on a path from the end system to any other distant end system. Examples of access networks are ISP, home networks, enterprise networks, ADSL, mobile networks, FTTH, etc.

Working Principle:

1. **Device Setup:** Users need a **modem, router, or switch** to connect to the internet.
2. **Connect to ISP:** The device uses **network protocols** to connect to the **ISP’s server** and start the internet connection.
3. **Choose Connection Type:** Depending on location and equipment, users can use **DSL, Ethernet, Wi-Fi, or 5G**.
4. **Data Transfer in Local Network:** Data moves between connected devices using **internal protocols** like **IP (Internet Protocol)**.
5. **Direct Device Communication:** Devices in the same network can communicate **without needing external routers**.
6. **Quality of Service (QoS):** The network can **prioritize important traffic** (like video calls) for **better performance**.

CT Types of access networks:

- **Ethernet** – It is the most commonly installed wired LAN technology, and it provides services on the **Physical and Data Link Layer** of OSI reference model. Ethernet LAN typically **uses coaxial cable or twisted pair wires**.
- **DSL** – DSL stands for **Digital Subscriber Line** and DSL brings a connection into your home through telephone lines and a DSL line can carry both **data and voice signals** and the data part of the line is continuously connected. In DSL you are able to use the **Internet and make phone calls simultaneously**. DSL modem uses the telephone lines to exchange data with digital subscriber line **access multiplexer (DSLAMs)**. In DSL we get **24 Mbps** downstream and **2.5 Mbps** upstream.
- **FTTH** – **Fiber to the home (FTTH)** uses optical fiber from a central Office directly to individual buildings and it provides high-speed Internet access among all access networks. It ensures **high initial investment but lesser future**

investment and it is the most expensive and most future-proof option amongst all these access networks.

- **Wireless LANs** – It links two or more devices using wireless communication within a range. It uses high-frequency radio waves and often include an access point for connecting to the Internet.
- **3G, LTE** – It uses cellular telephony to send or receive packets through a nearby base station operated by the cellular network provider. The term “3G internet” refers to the third generation of mobile phone standards as set by the International Telecommunications Union (ITU). Long Term Evolution (LTE) offers high-speed wireless communication for mobile devices and increased network capacity.
- **Satellite Internet** – Satellite internet is a wireless connection that uses satellite communication to deliver internet access to remote and rural areas. It has a higher latency and lower bandwidth compared to other access networks, but it can provide internet access in areas where other options are not available.
- **WiMAX** – WiMAX (Worldwide Interoperability for Microwave Access) is a wireless access network technology that provides high-speed internet access over a wide area. It is commonly used in rural and suburban areas where it is difficult or expensive to deploy wired networks.
- **5G** – 5G is the latest wireless communication technology that offers high-speed internet access and increased network capacity. It is designed to support a wide range of applications, including virtual and augmented reality, autonomous vehicles, and smart cities. 5G networks are being rolled out globally, and they are expected to transform the way we connect to the internet.
- **Wi-Fi:** Wi-Fi is a wireless access network technology that allows devices to connect to a local area network (LAN) or the Internet using radio waves. It is commonly used in homes, offices, public places, and other areas where people need wireless internet access.
- **Bluetooth:** Bluetooth is a short-range wireless communication technology that is used to connect devices within a limited range, typically up to 30 feet. It is commonly used to connect mobile phones, laptops, and other devices to speakers, headphones, and other accessories.

- **ZigBee:** ZigBee is a wireless communication technology that is used for low-power, low-speed applications such as home automation, industrial control, and sensor networks. It is designed to be simple, reliable, and easy to use.
- **LoRaWAN:** LoRaWAN is a wireless communication technology that is used for long-range, low-power applications such as smart cities, agriculture, and environmental monitoring. It is designed to be low-cost, low-power, and easy to deploy in remote or hard-to-reach areas.

Internet Applications

What Is VoIP?

Voice over Internet Protocol (**VoIP**) is a technology that allows you to make and receive phone calls over a broadband internet connection instead of a traditional analog phone line. This means no more complicated hardware or expensive long-distance charges.

VoIP is a popular choice for businesses because it offers significant cost savings, especially for long-distance and international calls. It also eliminates the need for extensive wiring, making it ideal for remote teams and flexible work arrangements.

How Does VoIP Work?

VoIP works by converting your voice (analog audio signals) into digital data packets, compressing them, and sending them over the internet rather than traditional phone systems or mobile networks.

VoIP systems bypass phone lines and use your existing internet infrastructure for making and receiving calls. The result is a flexible and cost-effective business phone solution.

VoIP Phone Service: Cloud-Based

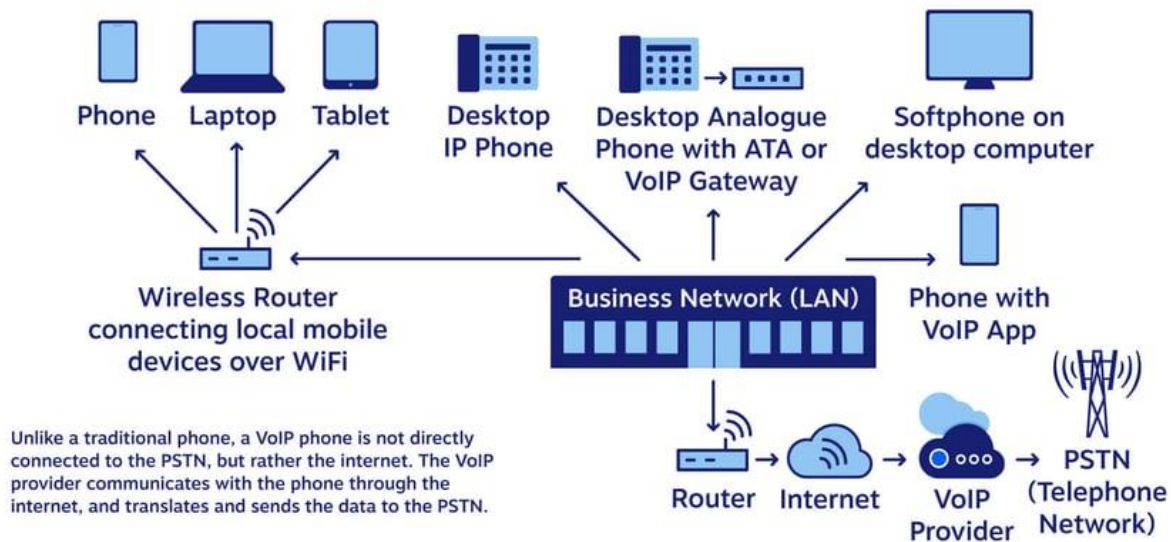


Fig: Network diagram of how VoIP works & transmits calls

VoIP might sound complex, but its underlying process is quite straightforward. Let's explore how VoIP works step by step:

1) Registration

Your VoIP phone or software establishes its presence with a VoIP server through a process called Session Initiation Protocol (**SIP**) registration running on an IP network. The server records your device's IP address and other relevant information, knowing that your device is online and ready to handle calls.

Once your device is online and ready to handle calls and when someone tries to call you, the server knows where to direct the call.

2) Establishing a connection

When you dial a number, your VoIP device sends a SIP INVITE request to the VoIP server. This request includes details about the caller and the intended recipient.

The server acts as an intermediary, locating the recipient's VoIP device (or their traditional phone if using an adapter) and setting up the connection between both parties.

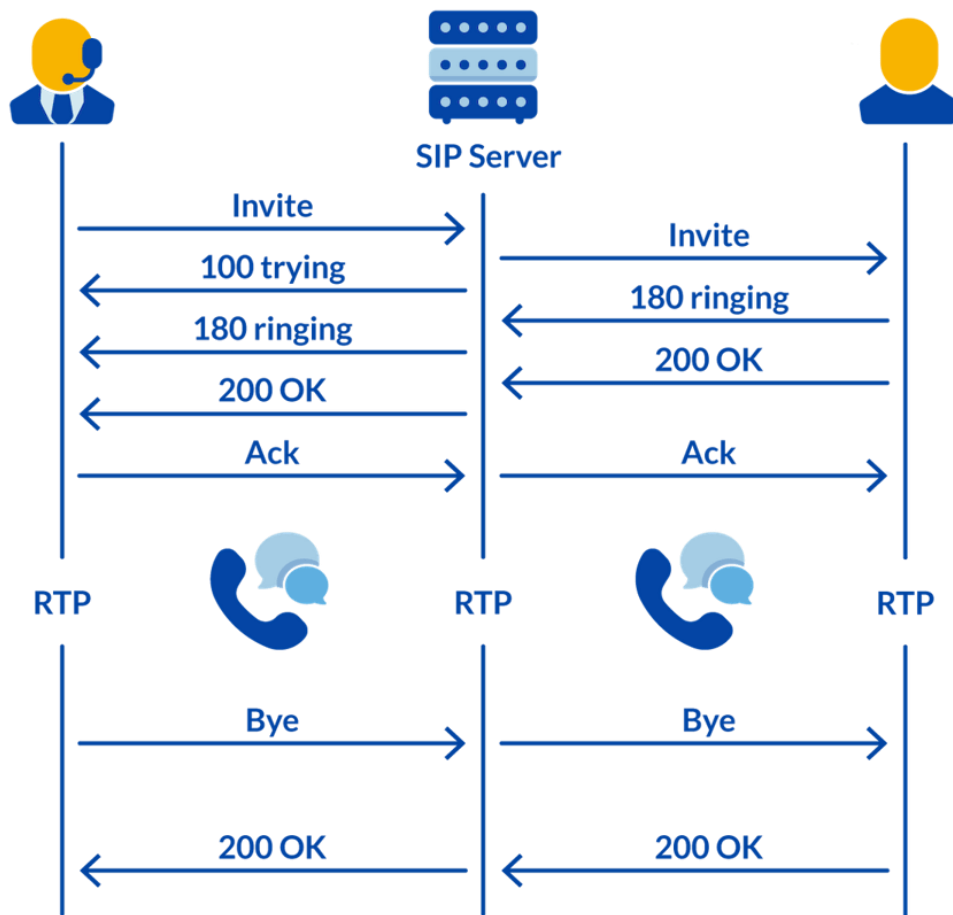


Fig: SIP Protocol Diagram

3) Sending and receiving voice data

As you speak, your voice is converted from analog sound waves into digital data by an Analog-to-Digital Converter (ADC). The digital data is then broken down into small packets.

These packets are transmitted over the internet using protocols like Real-time Transport Protocol (RTP) or User Datagram Protocol (UDP). On the receiving end, the packets are collected, reassembled in the correct order, and converted back into analog sound waves by a Digital-to-Analog Converter (DAC).

4) Ending the call

When the conversation is over, and either party hangs up, a SIP BYE message is sent to the VoIP server. The server then terminates the session, disconnects the devices, and frees up the resources used for the call.

It's good to know the inner workings of VoIP, but as a business, you essentially just need a reliable **VoIP phone service** that takes care of all the backend infrastructure, while you focus on connecting with your customers without thinking about the technicalities.